① Scalar components of $\vec{v}$ along $\vec{e}_1$ and $\vec{e}_2$ are

$$\vec{v}\cdot\vec{e}_1 = 2\left(\frac{1}{2}\right) + 7\left(\frac{\sqrt{3}}{4}\right) = \frac{4+7\sqrt{3}}{4}$$

$$\vec{v}\cdot\vec{e}_2 = 2\left(\frac{2}{3}\right) + 7\left(\frac{\sqrt{5}}{3}\right) = \frac{4+7\sqrt{5}}{4}$$

Vector components are →

$$(v.e_1)\, e_1 = \left\langle \frac{4+7\sqrt{3}}{4}\left(\frac{1}{2}\right), \frac{4+7\sqrt{3}}{4}\left(\frac{\sqrt{3}}{4}\right) \right\rangle$$

$$\Rightarrow \left\langle \frac{4+7\sqrt{3}}{8}, \frac{21+4\sqrt{3}}{16} \right\rangle$$

$$(v.e_2)\, e_2 = \left\langle \frac{4+7\sqrt{3}}{\cancel{4}_2}\left(\frac{\cancel{2}}{3}\right), \frac{4+7\sqrt{5}}{4}\left(\frac{\sqrt{5}}{3}\right) \right\rangle$$

$$= \left\langle \frac{4+7\sqrt{5}}{6}, \frac{35+4\sqrt{5}}{12} \right\rangle$$

(2) (a)

Here,

Line L_1 is parallel to vector $\vec{n}_1 = \langle 2, 1, 1 \rangle$

and,

Line L_2 is parallel to vector $\vec{n}_2 = \langle 8, 5, 2 \rangle$

Here,

$$\frac{2}{8} \neq \frac{1}{5} \neq \frac{1}{2}$$

$$\Rightarrow \frac{1}{4} \neq \frac{1}{5} \neq \frac{1}{2}$$

Therefore, $\vec{n}_1 \neq K \vec{n}_2$, so the line's are not parallel.

(b) Lets, assume

Line L_1 and line L_2 intersects

at the point (x_0, y_0, z_0). therefore,

$x_0 = 1 + 2t_1 = 2 + 8t_2$ ......(i)

$y_0 = 3 + t_1 = 2 + 5t_2$ ......(ii)

$z_0 = 1 + t_1 = 4 + 2t_2$ ......(iii)

from (i) and (ii) $\rightarrow$

$1 + 2t_1 = 2 + 8t_2$

$t_1 - 5t_2 = -1$

$2t_1 - 8t_2 = 1$

$t_1 - 5t_2 = -1$

$t_1 = \frac{13}{2}$, $t_2 = \frac{3}{2}$

so,
the Line will intersect if and only if the valus of
t_1 and t_2 satisfy the equation (iii)

from (iii) $\rightarrow$

$1+t_1 = 4+2t_2$

L.H.S	R.H.S
$1+\frac{13}{2}$	$4+\cancel{2}\cdot\frac{13}{\cancel{2}}$
$\Rightarrow \frac{15}{2}$	$\Rightarrow 17$

There fore L.H.S $\neq$ R.H.S so, line L_1 and Line L_2
do not intersects

(c) Here,

$\vec{n}_1 = \langle 2, 1, 1 \rangle$

and $\vec{n}_2 = \langle 8, 5, 2 \rangle$

Therefore the normal vector $\rightarrow$

$\vec{n} = \vec{n}_1 \times \vec{n}_2$

$\Rightarrow \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & 1 \\ 8 & 5 & 2 \end{vmatrix}$

$\Rightarrow -3\hat{i} + 4\hat{j} + 2\hat{k}$

$\Rightarrow \langle -3, 4, 2 \rangle$

There for the equation of the plane $\rightarrow$

$$-3(x-1)+4(y-3)+2(z-1)=0$$

$$\Rightarrow -3x+3+4y-12+2z-2=0$$

$$\Rightarrow -3x+4y+2z-11=0$$

Here,

point$_1$ $(1,3,1)$

point$_2$ $(2,8,4)$

So, the equation of plane whic contains L_1 is

$$-3x+4y+2z-11=0$$

agin the point$_2$ $\rightarrow$ $(2,8,4)$

if the Distance detween L_1 & L_2 is D then

$$D=\frac{|(-3)(2)+(4)(8)+2(4)-11|}{\sqrt{(-3)^2+(4)^2+(2)^2}}$$

$$=\frac{|23|}{\sqrt{29}}$$

$$\approx 4.27 \quad A$$

③ if the parallel vector of the line, $\vec{v} = \langle -1, 1, 1 \rangle$ and the normal vector of the plane $\vec{n} = \langle 2, 1, 1 \rangle$ are orthogonal

then we can call the line $x = 2-t$, $y = 3+t$, $z = t$ is parallel to the plane $2x + y + z = 1$.

Now,

$\vec{v} \cdot \vec{n} = (-1)(2) + (1)(1) + (1)(1)$

$\Rightarrow -2 + 1 + 1$

$= 0$

Therefore, the line is parallel to the plane.

(4) we have,

$\vec{v} \cdot \vec{b} = \langle -3, -2 \rangle \cdot \langle 2, 1 \rangle$

$\Rightarrow -6 - 2 = -8$

Here,

$\vec{b} = 2\hat{i} + \hat{j}$

$\vec{v} = -3\hat{i} - 2\hat{j}$

again

$\|\vec{b}\|^2 = 2^2 + 1^2$

$= 5$

Thus the projection of v on b is

$\text{proj}_b v = \frac{v \cdot b}{\|b\|^2} b$

$= -\frac{8}{5}(2\hat{i} + \hat{j})$

$\Rightarrow -\frac{16}{5}\hat{i} - \frac{8}{5}\hat{j}$ A

the vector component of v orthogonal to b →

$v - \text{proj}_b v = (-3\hat{i} - 2\hat{j}) - \left(-\frac{16}{5}\hat{i} - \frac{8}{5}\hat{j}\right)$

$= \frac{1}{5}\hat{i} - \frac{2}{5}\hat{j}$

(5) Since the points P_1, Q_2 and R_3 lie in the plane the vectors $\overrightarrow{PQ} = \langle 3, 3, -8 \rangle$ and $\overrightarrow{PR} = \langle 3, -1, -3 \rangle$ are parallel to the plane. Therefore

$$\overrightarrow{P_1Q} \times \overrightarrow{P_1R_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 3 & -8 \\ 3 & -1 & -3 \end{vmatrix}$$

$$d = -17\,\hat{i} - 15\,\hat{j} - 12\,\hat{k}$$

is the plane, and orthogonal to both $\overrightarrow{PQ}$ and $\overrightarrow{PQ}$. By using point $P(-1, 4, 8)$ ~~(10, 2, -0)~~ in the plane we get →

$$-17(x+1) - 15(y-4) - 12(z-8) = 0$$

$$\Rightarrow -17x - 17 - 15y + 60 - 12z + 96 = 0$$

$$-17x - 15y - 12z + 139 = 0$$

⑥ Here,

the dimensions of the box are 10 cm, 15 cm, and 25 cm

therefore, diagonal vector $\vec{d} = 10\hat{i} + 15\hat{j} + 25\hat{k}$

$$\therefore \|\vec{d}\| = \sqrt{(10)^2 + (15)^2 + (25)^2}$$

$$= \sqrt{950} \approx 30.81\text{ cm}$$

Therefore,

Angle with the x axis $\alpha \rightarrow$

$$\alpha = \cos^{-1}\left(\frac{10}{30.81}\right)$$

$$\approx 71^\circ$$

Angle with the y axis $\beta \longrightarrow$

$$\beta = \cos^{-1}\left(\frac{15}{30.81}\right)$$

$$\approx 61^\circ$$

Angle with the z-axis $\gamma \rightarrow$

$$\gamma = \cos^{-1}\left(\frac{25}{30.81}\right)$$

$$\simeq 36^\circ$$

So, The angles that the diagonal makes with the edges of the box are

$$\alpha \simeq 71^\circ, \quad \beta = 61^\circ, \quad \gamma = 36^\circ$$

7 Let,

$$\vec{a} = \langle 4, -8, 1 \rangle$$

$$\vec{b} = \langle 2, 1, -2 \rangle$$

$$\vec{c} = \langle 3, -4, 12 \rangle$$

We know,

the volume of the parallelepiped, $V = |\vec{a} \cdot (\vec{b} \times \vec{c})|$

Now,

$$\vec{b} \times \vec{c} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 3 & -4 & 12 \end{vmatrix} = 4\hat{i} - 30\hat{j} - 11\hat{k}$$

$$= \langle 4, -30, -11 \rangle$$

Therefore,

$$V = \langle 4, -8, 1 \rangle \cdot \langle 4, -30, -11 \rangle$$

$$= 16 + 240 - 11$$

$$= 245$$ Ans